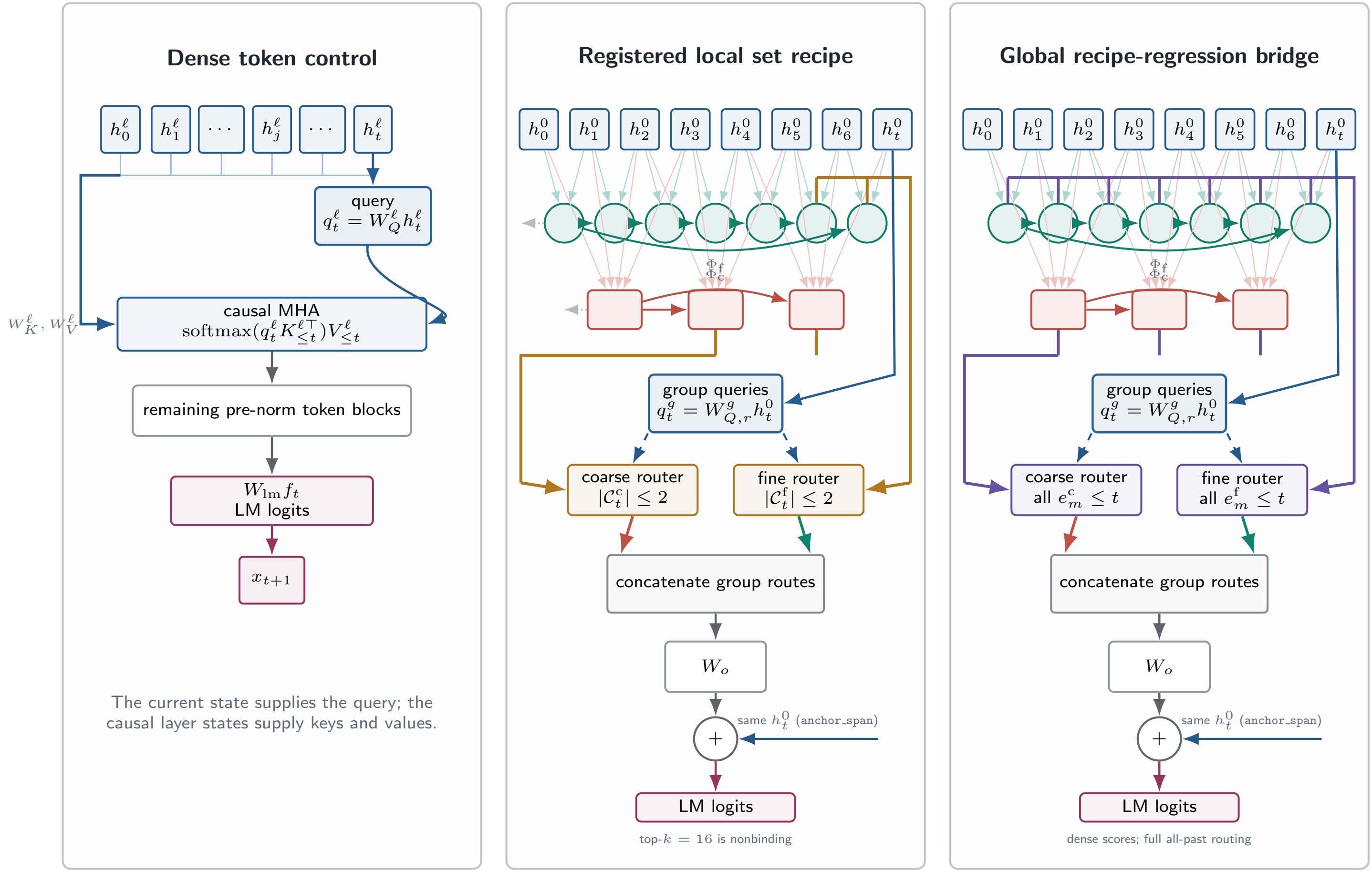
WIKITEXT-2: LOCAL AND GLOBAL CONTEXT PATHS

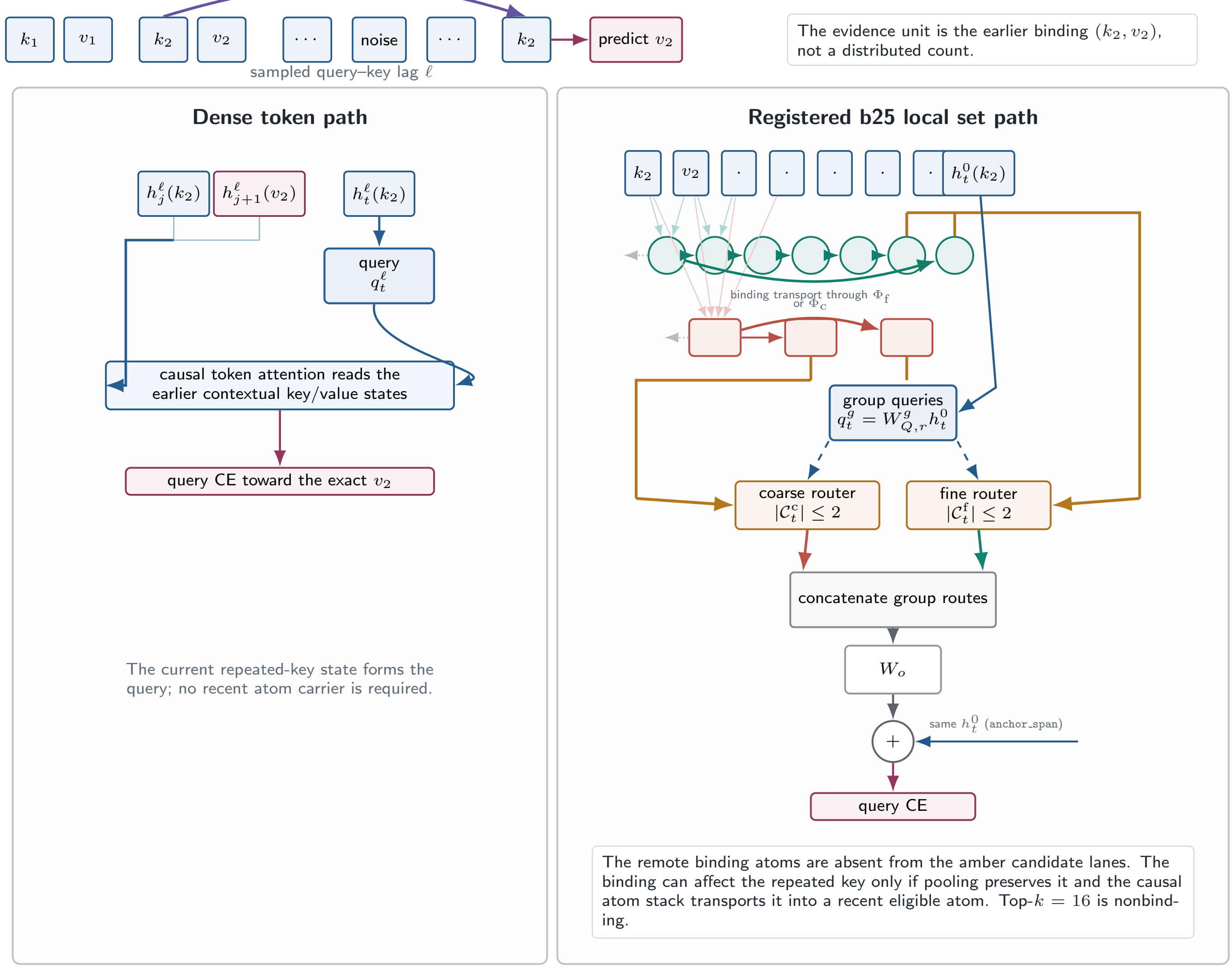
Every non-masked position predicts the next token; each column follows one executed model from states to logits.



Shared visual grammar. Every computation runs downward. Upper green circles are fine atoms; lower red rectangles are coarse atoms. A blue query box is a learned projection of the current state, never a merge or decision. Candidate rails lie outside the atom rows: fine above, coarse below. Amber rails admit recent endpoints; violet rails admit every sealed past atom. Rails identify eligible indices: router keys use pre-stack pooled states and routed values use post-stack contextualized atoms.

SYNTHETIC MQAR: EXACT BINDING TRANSPORT

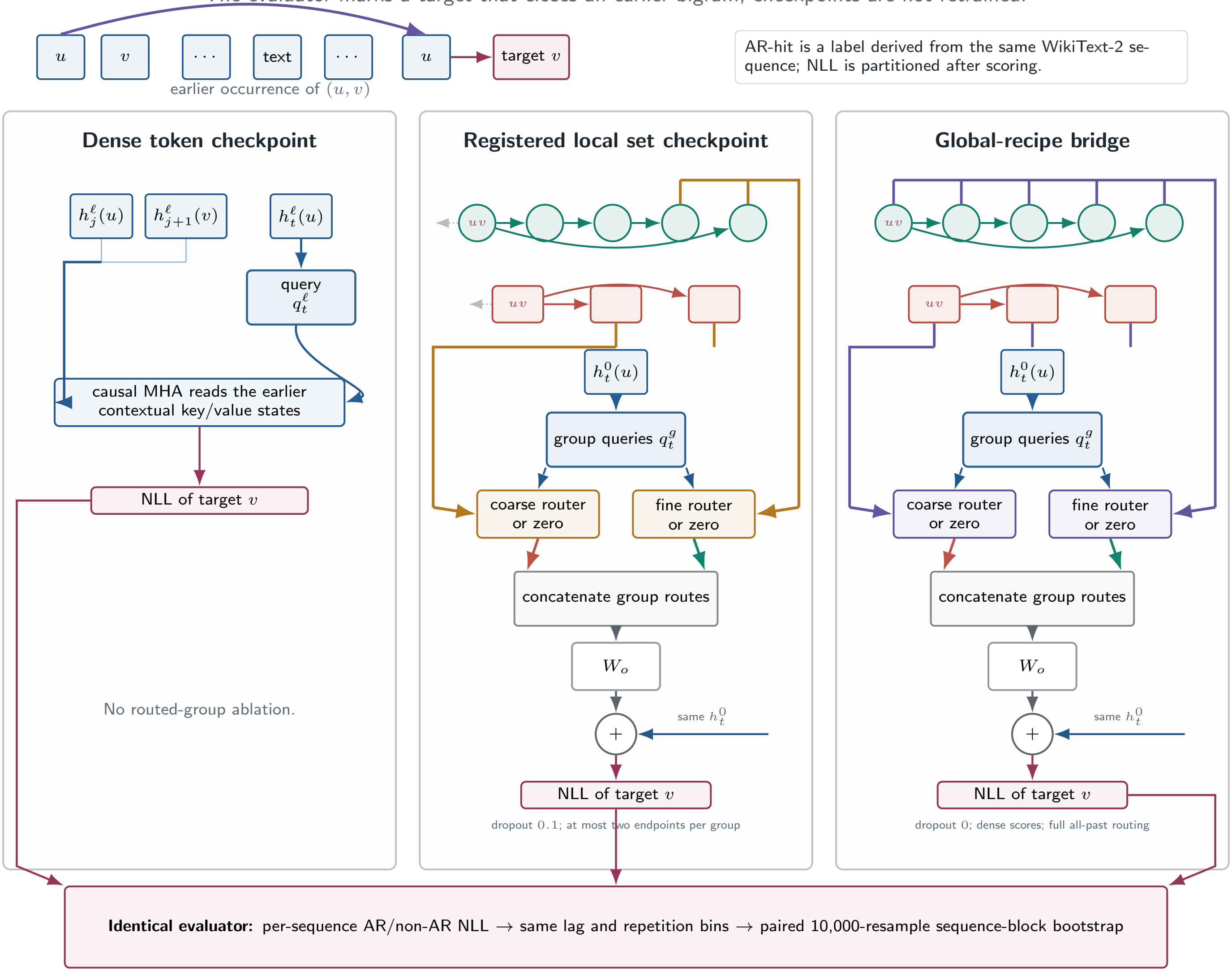
A repeated key must recover its earlier paired value; only query-target positions are supervised.



This result tests local-recipe binding transport; it does not substitute the later all-past/full-routing recipe.

NATURAL REPEATED-BIGRAM AR: FIXED TARGET, DIFFERENT READBACK

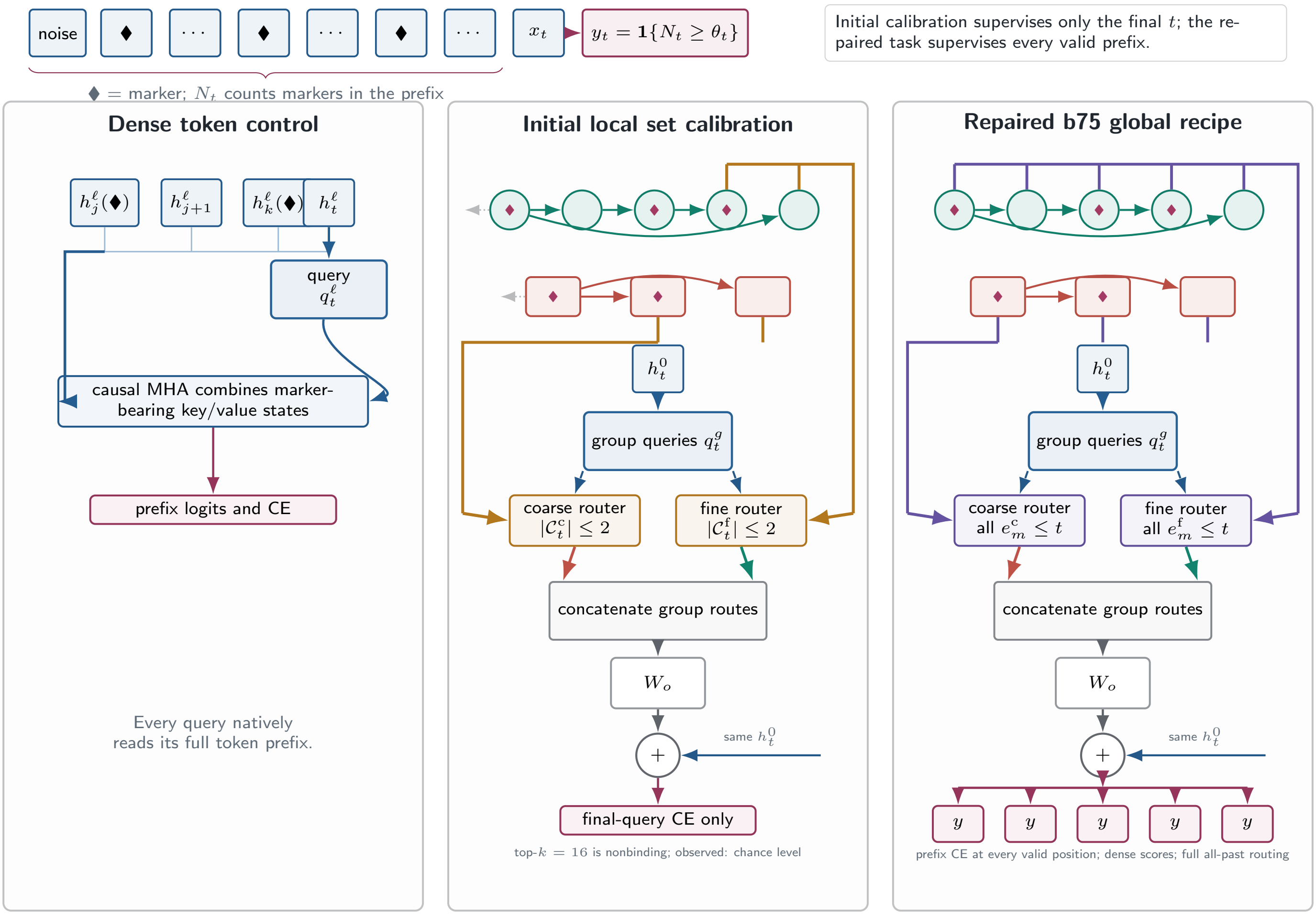
The evaluator marks a target that closes an earlier bigram; checkpoints are not retrained.



Only the contextual path changes. Group removal measures dependence on an allocated fine/coarse route, not a dedicated retrieval circuit.

LCA MARKER COUNTING: DISTRIBUTED EVIDENCE AND READBACK

For each supervised prefix, predict whether its marker count exceeds the configured threshold.



Three controlled changes. Candidate reachability changes from two recent endpoints to every sealed past atom; routing bandwidth changes from an already nonbinding top-k = 16 to full routing; supervision changes from one final target to every valid prefix. Score mode changes memory materialization, not the dot-product/softmax routing equation.